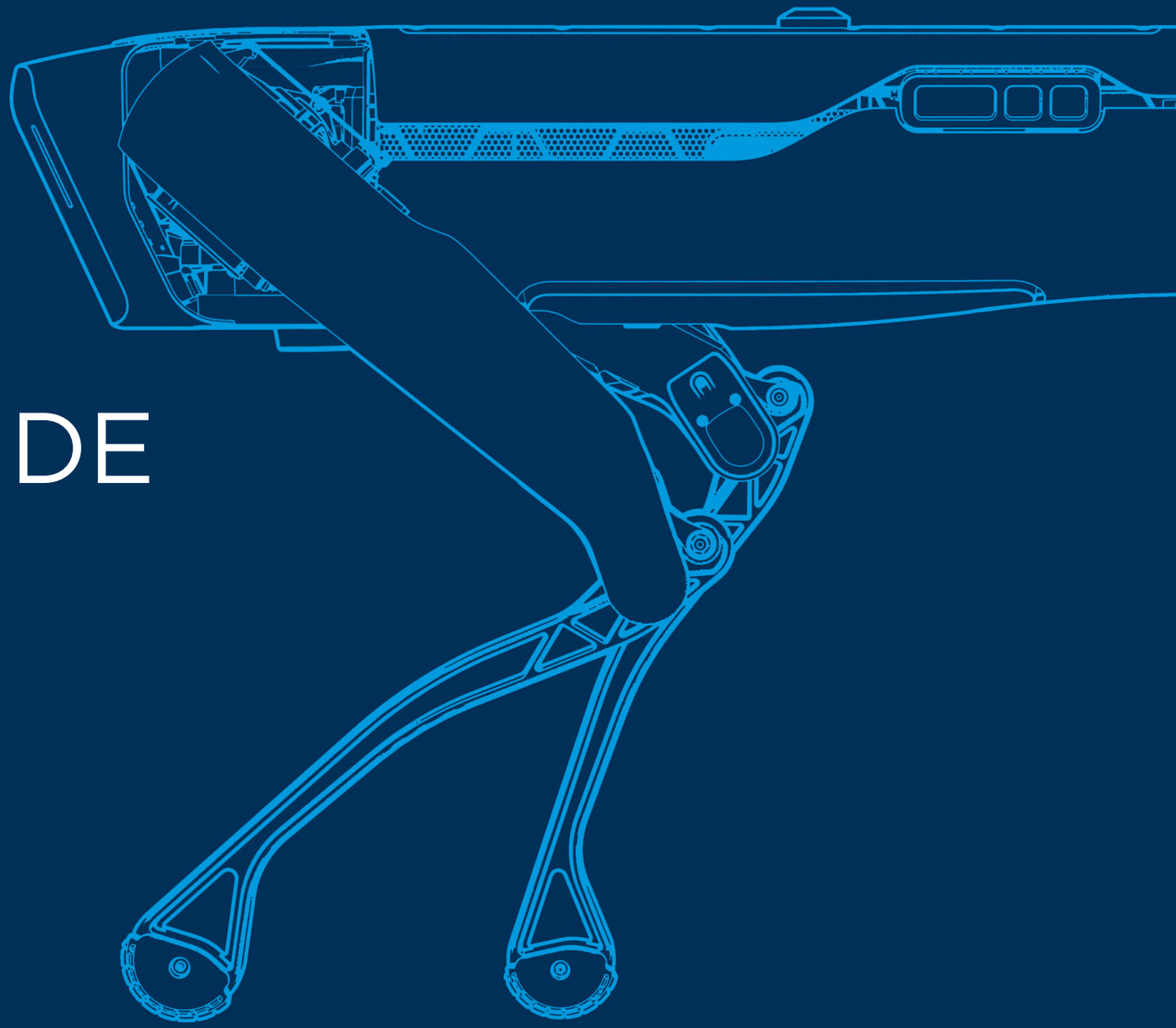
Boston Dynamics

SPOT

USER GUIDE

RELEASE 1.1 VERSION A



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SPECIFICATIONS

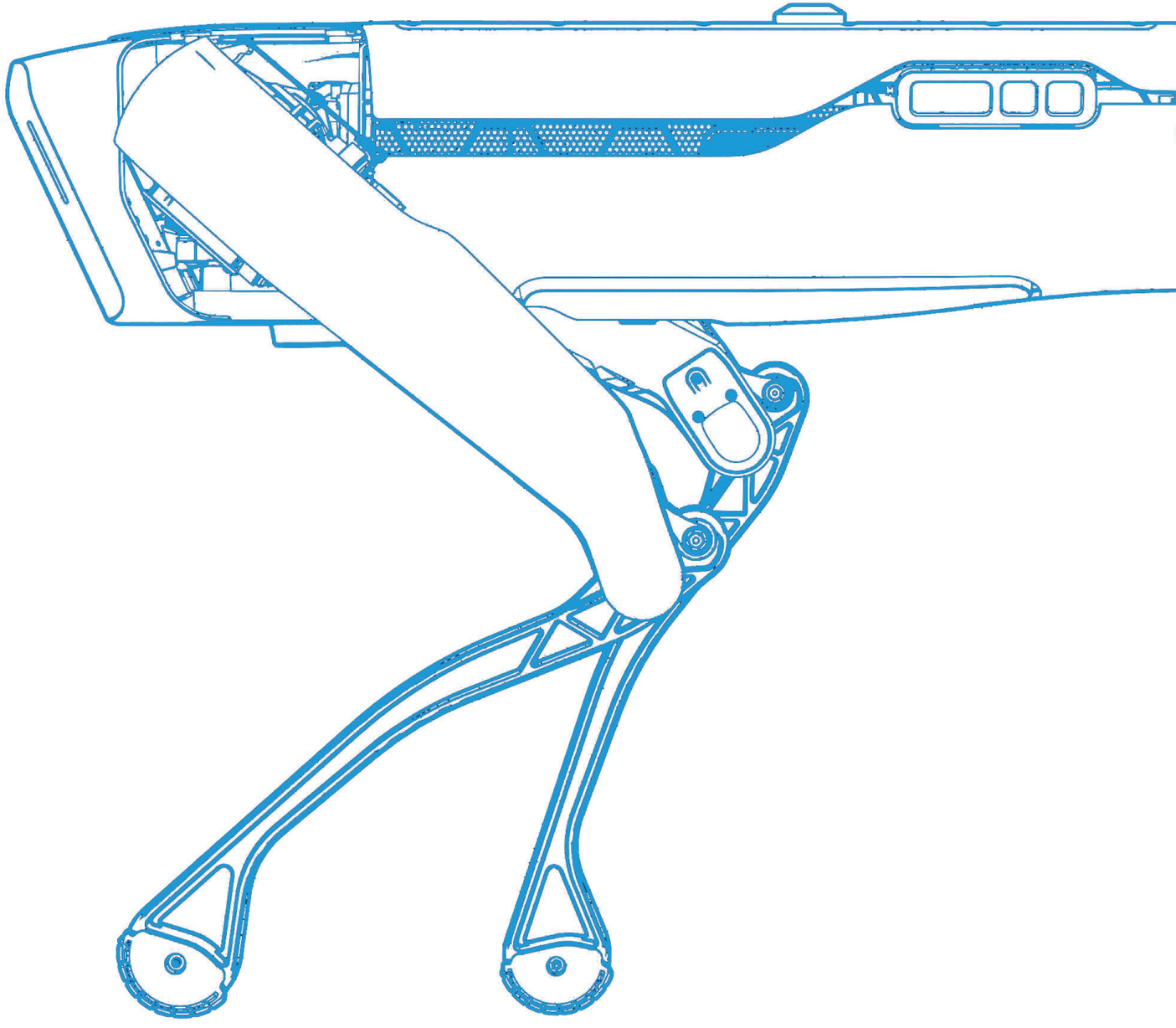
SPECIFICATIONS **61**

LEGEND

WARNING: Indicates a hazardous situation which, if not avoided, could result in death or serious injury. Such situation could also result in significant damage to the robot or property.

CAUTION: Indicates a hazardous situation which, if not avoided, could result in minor to moderate injury. Such situation could also result in minor to moderate damage to the robot or property.

NOTE: Addresses practices not related to safety, including tips to maximize performance and prevent damage to the robot



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INTRODUCING SPOT

R 1.1
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01

HELLO

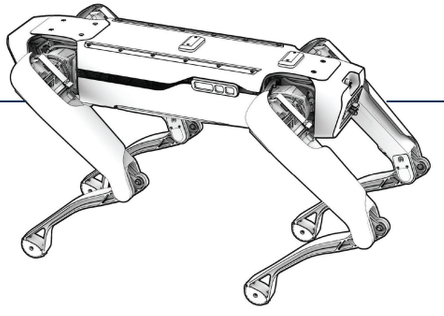
We’ve built the world’s most advanced four-legged robot. Spot is small enough to navigate human environments, but nimble enough to traverse nearly any terrain. Spot dynamically balances, but is easy to control. As Spot takes its first steps out of the lab, it represents a giant leap toward a future of general-purpose robotics.

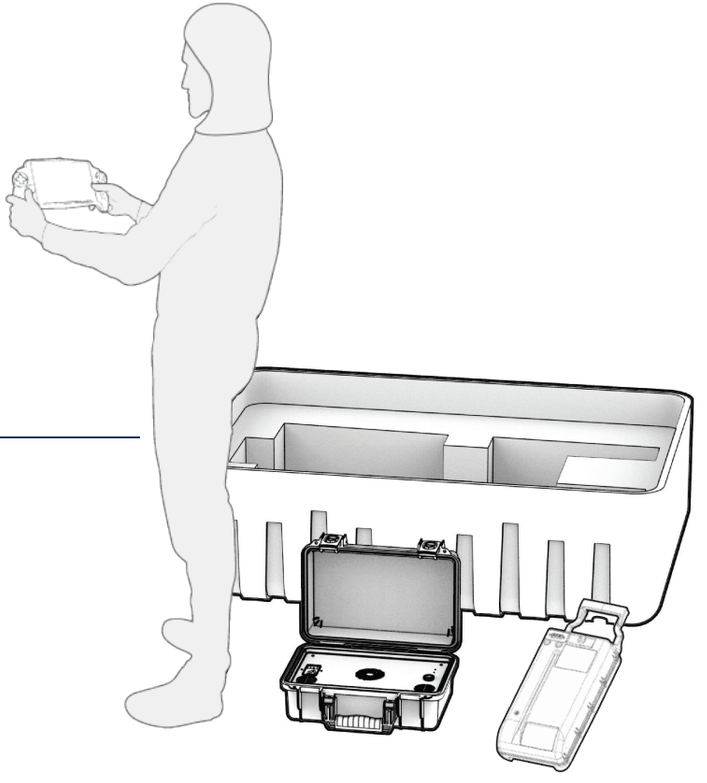
Robotics is a team sport. Boston Dynamics has built the platform – and it’s up to you to put it to work. With this user guide, a growing list of payloads, evolving software capabilities, and accessible development kits, you have the tools to leverage this technology to its fullest.

It’s time to change the world’s idea of what robots can do.
Let’s get started.

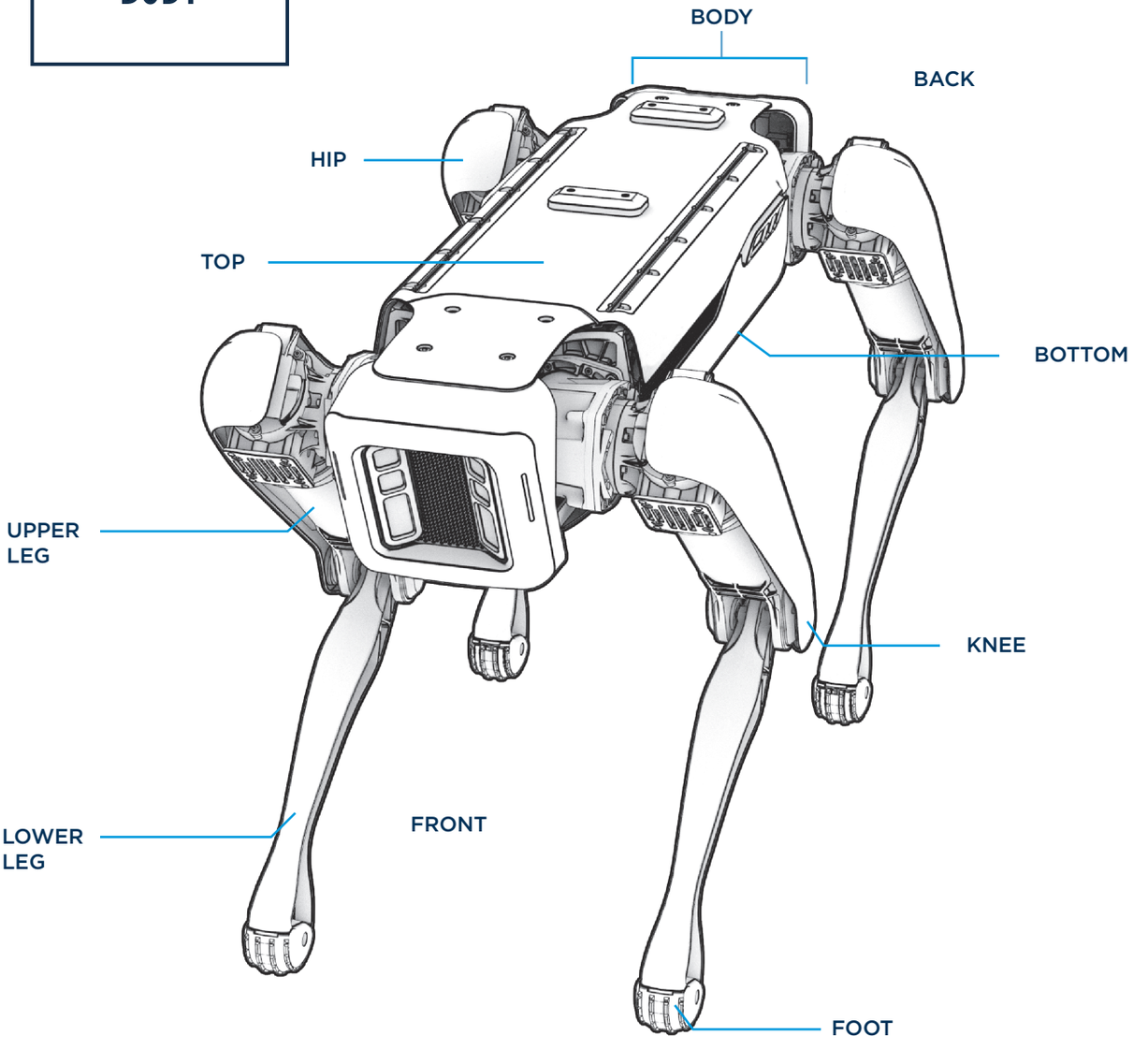
SYSTEM CONTENTS

- Spot Robot
- Battery (x2)
- Charger
- Controller



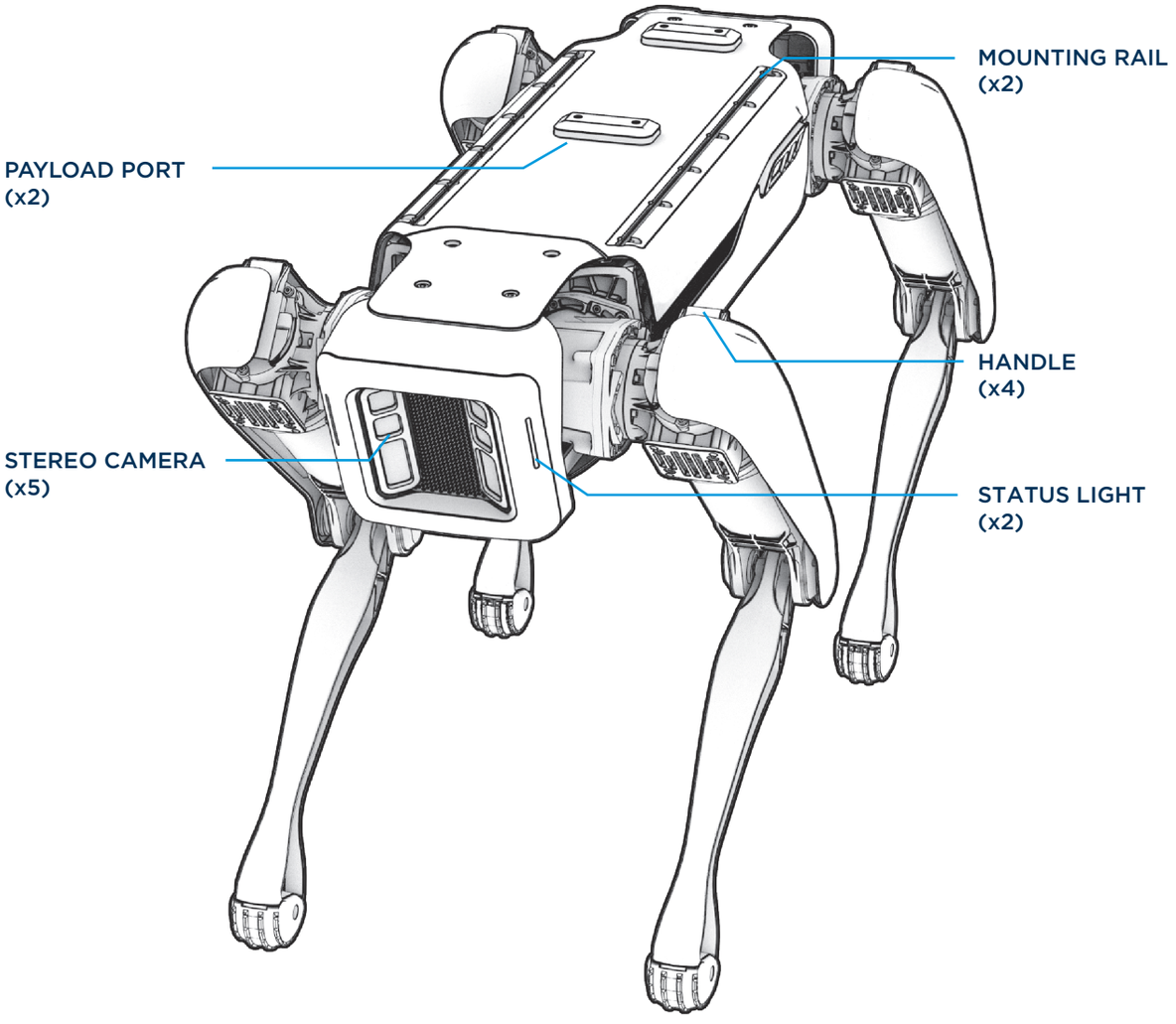


BODY



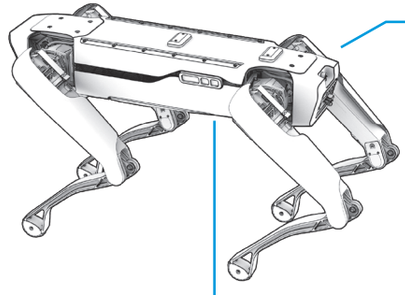
THREE-QUARTER VIEW

INTRODUCING SPOT

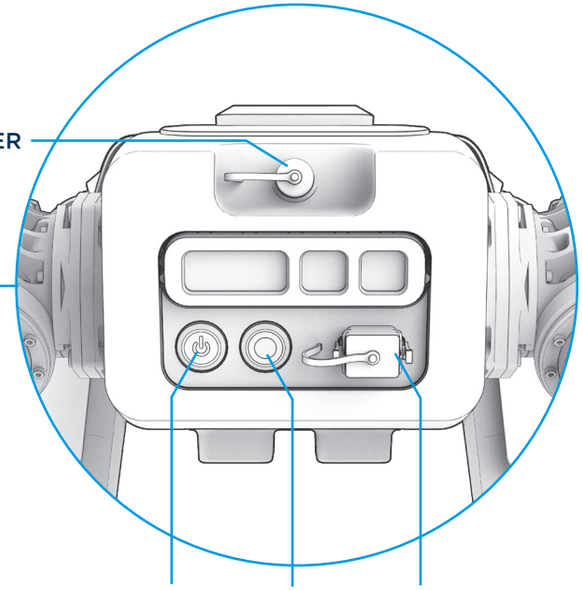


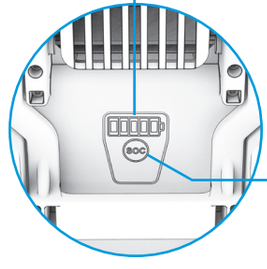
THREE-QUARTER VIEW

BUTTONS & PORTS



BACK VIEW



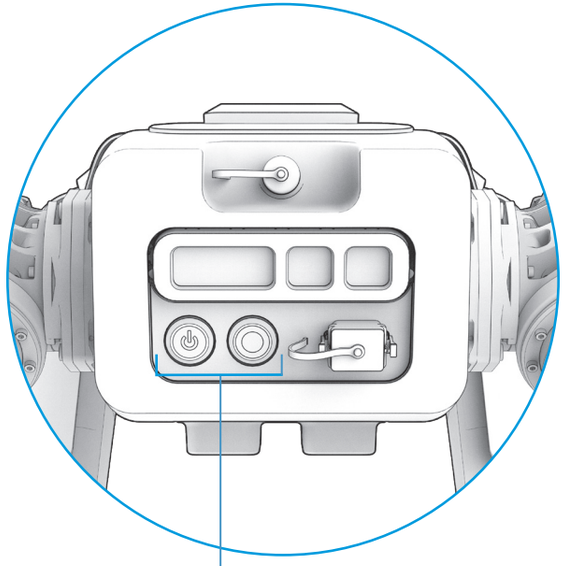


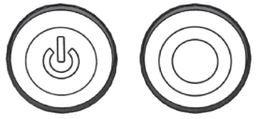
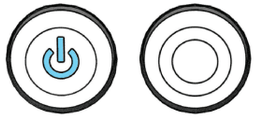
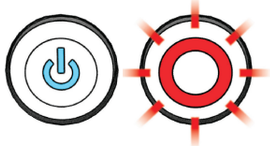
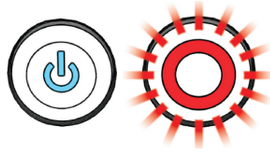
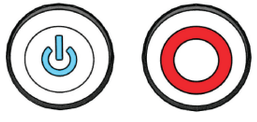
STATE OF CHARGE (SOC) BUTTON
Shows the battery charge level; located on the battery

BOTTOM VIEW

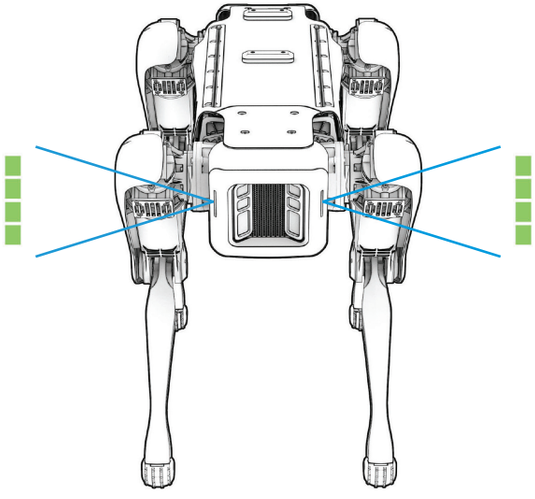
POWER & MOTOR LIGHTS

BACK VIEW



POWER BUTTON	LOCKOUT BUTTON	ROBOT STATUS
		**NO LIGHTS:** Robot and motors off (lockout engaged)
		**BLUE LIGHT**: Computers on, motors off (lockout engaged)
		**SLOWLY FLASHING RED LIGHT**: Motors available but not powered
		**QUICKLY FLASHING RED LIGHT**: Motors turning on
		**SOLID RED LIGHT:** Motors on - stay at least 2m away

STATUS LIGHTS



POWER

Yellow:
System booting up

Moving rainbow:
Powered on, ready to connect

Solid rainbow:
Plugged in, no battery

Rainbow dots:
Charging, dots show % charge

MOTORS

Solid blue:

Lockout button engaged;
Spot is safe to handle

Slow blink green:
Spot connected; motors may turn on

Fast blink green:
Motors turning on!

Solid green:
Motors on

ERRORS

Blinking orange:
Serious error or perception fault
has occurred; operation degraded

CONTROLLER

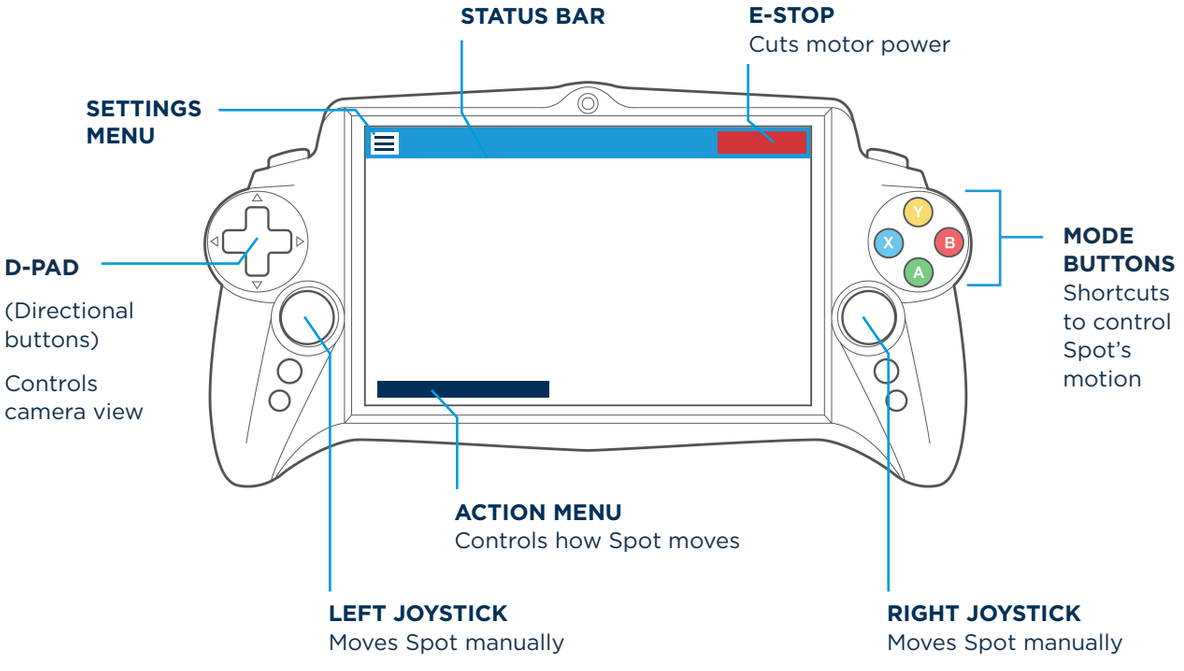
Spot is controlled by an android app running on a tablet. The Spot System includes an Android gaming tablet with physical buttons for easy robot control.

OPERATING MODES

Spot's controller has two operating modes. Change between the modes in the settings menu.

Standard Mode: Provides easy-to-use control for most situations.

Demo Mode: Enables demonstration gaits and adds additional button controls to make it easy to switch modes.



TRAINING

Proper training: Do not use Spot without reading this user guide and understanding safe operation of the robot.

Responsible use of Spot is crucial to prevent dangerous conditions for operators and others nearby. Make sure to read, understand and comply with this user guide to lessen the risk of serious injury, death, or robot and other property damage.

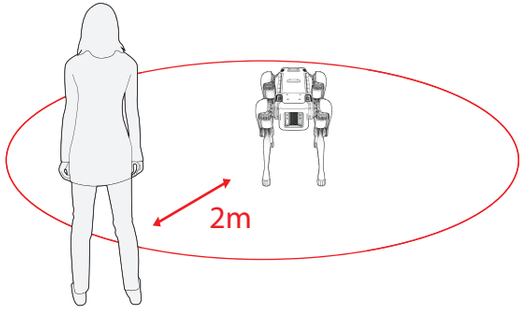
It is the operator's responsibility to ensure that **bystanders are trained in safe behavior around Spot**, and that the operation of Spot does not put anyone at risk.

SAFETY DISTANCE

Spot is not suitable for tasks that require operation in close proximity to people. **People must stay a safe distance (at least 2m) from Spot during operation to avoid injury.** Injuries may be caused by collisions, Spot falling or tipping onto people, or contact with Spot's pinch points.

Spot is suitable for areas where access is limited to trained personnel. Keep untrained people away from Spot during operation to avoid injury.

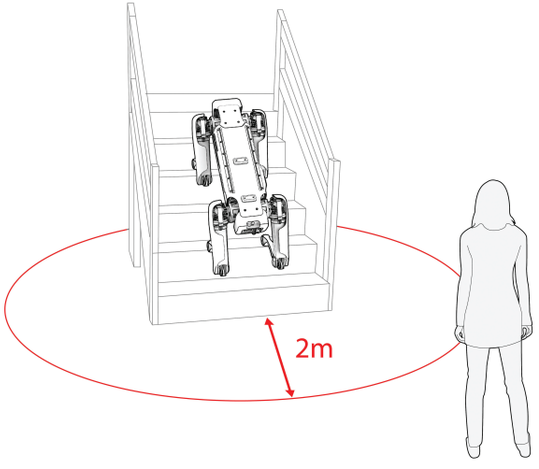
Spot may sometimes move unpredictably or fall. Only use Spot in areas where a fall or collision will not result in an unacceptable risk.



COLLISIONS

Spot can collide with people or objects, even with its obstacle detection system enabled.

- People around Spot should keep a safe distance of at least 2m from the robot while it is operating on level ground. Do not stand between Spot and a wall or other fixed object if this could violate the 2m safety perimeter.
- Do not operate Spot within 2m of delicate equipment or other property that may be damaged by the robot. Operators and anyone nearby should assume that Spot may move unexpectedly at any time.



FALLS

Spot may sometimes fall. Falls can damage Spot or the surrounding environment and injure operators and bystanders. Keep a safe distance of at least 2m from Spot while it is operating.

Never stand below Spot on stairways, elevated platforms, or inclines. For example, if Spot is at the top of a stairway, stay at least 2m away from the stairway bottom or any open areas on the side of the stairway and remain vigilant at all times to avoid a falling Spot. If Spot falls from an elevated position, it can cause serious injury or death.

The following conditions may cause Spot to fall:

Stairs or inclines	If Spot loses balance on the stairs or an incline, it may slip and fall.
Holes, elevated grounds, or cliffs	Spot cannot detect cliff edges and may walk off cliff edges that are unblocked. Block cliff edges with secure guardrails or safety netting. Never work below an elevated Spot.
Signal loss	If Spot loses its wireless connection to the controller, it will sit after 3 seconds, and the motors will shut off after 8 seconds. If this occurs on stairs, slope, or other uneven surface, Spot may slide or fall upon attempting to sit.
Slippery surfaces	Spot has difficulty walking, and may fall, on slippery surfaces. This includes surfaces made slippery by sand, dust, and liquids, for example.
Cords	Spot cannot detect when its legs are entangled in cords, which may cause it to trip and fall. The entagled cord and anything connected to it may also be damaged.
Unstable objects shorter than 30 cm	Spot may not automatically avoid unstable objects that are fully or partially less than 30 cm above the ground, such as human feet and backpacks, and may fall upon stepping on such objects.
Transparent, mirrored, or very bright obstacles	These can confuse Spot's obstacle detection system.
Moving surfaces	Spot can become unstable on moving surfaces, such as elevators, escalators, moving walkways and ramps, vehicles, or boats. Put robot in lockout mode when transporting on moving platform.

SAFETY IN HANDLING

PINCH POINTS

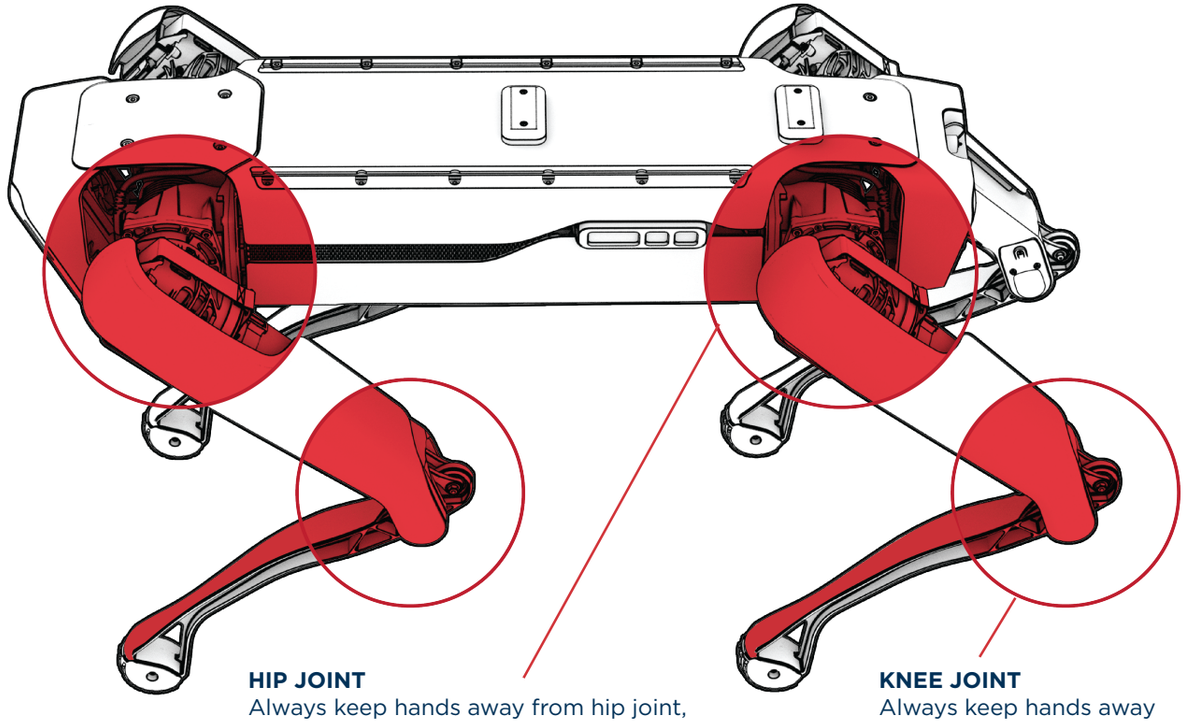
Spot's joints can pinch fingers and other body parts and entangle loose clothing, long hair, and jewelry.

Only handle Spot when the motors are locked out (see page 14) or robot power is off.

ACTIVE MOTORS

Fingers may break or get amputated if caught in joints while Spot's motors are active. Stay at least 2m away when Spot is powered on (except to press the lockout or power buttons).

1. Never touch Spot when motors are active. Stay at least 2m away at all times.
2. Before handling Spot, engage lockout to ensure motors can't turn on.



HIP JOINT
Always keep hands away from hip joint, except when grabbing handles. Always make a fist when gripping handles to keep your fingers away from pinch points.

KNEE JOINT
Always keep hands away from knee joint. Use caution when lowering Spot or closing the leg.